

TP PANDAS PARTIE 2 Realisé par : Salma El Angui

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In [ ]: import pandas as pd
import pandas as pd

url = "https://raw.githubusercontent.com/justmarkham/DAT8/master/data/chipotle.txt"

df = pd.read_csv(url, sep='|')
print(df.head())
```

```
In [4]: #Affecter le jeu de données à une variable appelée chipo.
chipo=pd.read_csv(url, sep='|')
```

```
In [20]: #Afficher les 10 premières et les 10 dernières entrées du DataFrame.
top_10_lines=df.head(10)
print(top_10_lines)
#solution2
top_10_lines=df.iloc[: 10]
print(top_10_lines)
```

	order_id	quantity	item_name \
0	1	1	Chips and Fresh Tomato Salsa
1	1	1	Izze
2	1	1	Nantucket Nectar
3	1	1	Chips and Tomatillo-Green Chili Salsa
4	2	2	Chicken Bowl
5	3	1	Chicken Bowl
6	3	1	Side of Chips
7	4	1	Steak Burrito
8	4	1	Steak Soft Tacos
9	5	1	Steak Burrito

	choice_description	item_price
0	NaN	\$2.39
1	[Clementine]	\$3.39
2	[Apple]	\$3.39
3	NaN	\$2.39
4	[Tomatillo-Red Chili Salsa (Hot), [Black Beans...	\$16.98
5	[Fresh Tomato Salsa (Mild), [Rice, Cheese, Sou...	\$10.98
6	NaN	\$1.69
7	[Tomatillo Red Chili Salsa, [Fajita Vegetables...	\$11.75
8	[Tomatillo Green Chili Salsa, [Pinto Beans, Ch...	\$9.25
9	[Fresh Tomato Salsa, [Rice, Black Beans, Pinto...	\$9.25

	order_id	quantity	item_name \
0	1	1	Chips and Fresh Tomato Salsa
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9	[Fresh Tomato Salsa, [Rice, Black Beans, Pinto...	\$9.25

```
In [19]: last_10_lines=df.tail(10)
print(last_10_lines)
#solution 2
last_10_lines=df.iloc[-10 :]
print(last_10_lines)
```

	order_id	quantity	item_name \
4612	1831	1	Carnitas Bowl
4613	1831	1	Chips
4614	1831	1	Bottled Water
4615	1832	1	Chicken Soft Tacos
4616	1832	1	Chips and Guacamole
4617	1833	1	Steak Burrito
4618	1833	1	Steak Burrito
4619	1834	1	Chicken Salad Bowl
4620	1834	1	Chicken Salad Bowl
4621	1834	1	Chicken Salad Bowl

	choice_description	item_price
4612	[Fresh Tomato Salsa, [Fajita Vegetables, Rice,...	\$9.25
4613	NaN	\$2.15
4614	NaN	\$1.50
4615	[Fresh Tomato Salsa, [Rice, Cheese, Sour Cream]]	\$8.75
4616	NaN	\$4.45
4617	[Fresh Tomato Salsa, [Rice, Black Beans, Sour ...	\$11.75
4618	[Fresh Tomato Salsa, [Rice, Sour Cream, Cheese...	\$11.75
4619	[Fresh Tomato Salsa, [Fajita Vegetables, Pinto...	\$11.25
4620	[Fresh Tomato Salsa, [Fajita Vegetables, Lettu...	\$8.75
4621	[Fresh Tomato Salsa, [Fajita Vegetables, Pinto...	\$8.75

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4614	1831	1	Bottled Water
4615	1832	1	Chicken Soft Tacos
4616	1832	1	Chips and Guacamole
4617	1833	1	Steak Burrito
4618	1833	1	Steak Burrito
4619	1834	1	Chicken Salad Bowl
4620	1834	1	Chicken Salad Bowl
4621	1834	1	Chicken Salad Bowl

	choice_description	item_price
4612	[Fresh Tomato Salsa, [Fajita Vegetables, Rice,...	\$9.25
4613	NaN	\$2.15
4614	NaN	\$1.50
4615	[Fresh Tomato Salsa, [Rice, Cheese, Sour Cream]]	\$8.75
4616	NaN	\$4.45
4617	[Fresh Tomato Salsa, [Rice, Black Beans, Sour ...	\$11.75
4618	[Fresh Tomato Salsa, [Rice, Sour Cream, Cheese...	\$11.75
4619	[Fresh Tomato Salsa, [Fajita Vegetables, Pinto...	\$11.25
4620	[Fresh Tomato Salsa, [Fajita Vegetables, Lettu...	\$8.75
4621	[Fresh Tomato Salsa, [Fajita Vegetables, Pinto...	\$8.75

```
In [25]: #5. Quel est Le nombre d'observations dans Le jeu de données ?
#solution 1
len(chipo)
#solution 2
chipo.shape
#solution3
chipo.size
#solution4
chipo.ndim
```

```
Out[25]: 2
```

```
In [24]: #Quel est Le nombre de colonnes dans Le jeu de données ?
#solution1
chipo.shape[1]
#solution2
chipo.columns.size
```

Out[24]: 5

```
In [27]: #Afficher Le nom de toutes Les colonnes.
#solution1
print(df.columns)
#solution2
chipo=df[['order_id', 'quantity', 'item_name', 'choice_description',
          'item_price']]
```

```
Index(['order_id', 'quantity', 'item_name', 'choice_description',
       'item_price'],
      dtype='object')
```

```
In [28]: # 8. Comment Le jeu de données est-il indexé ?
# solution1
chipo.index
```

Out[28]: RangeIndex(start=0, stop=4622, step=1)

```
In [30]: #9Combien d'articles ont été commandés au total ?
# solution1
total_articles=chipo['quantity'].sum()
print(total_articles)
```

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```
In [31]: #10Convertir Le prix de L'article en nombre décimal (float)
#10.a Vérifier Le type de La variable item price. Expliquer Le resultat ?
chipo['item_price'].dtype
#Le résultat est object → ça signifie que Python considère La colonne comme du te
```

Out[31]: dtype('O')

```
In [32]: chipo['item_price']=chipo['item_price'].str.replace('$', '').astype(float)
chipo['item_price'].dtype
```

Out[32]: dtype('float64')

```
In [35]: #10.b Créer une fonction lambda et changer Le type de item_price (utiliser La me
chipo['item_price'] = chipo['item_price'].apply(lambda x: float(str(x).replace('
# 10.c. Vérifier Le type de La variable item price
chipo['item_price'].dtype
```

Out[35]: dtype('float64')

```
In [39]: # 11. Afficher uniquement La colonne quantity
chipo.quantity
```

```
Out[39]: 0      1
          1      1
          2      1
          3      1
          4      2
          ..
         4617    1
         4618    1
         4619    1
         4620    1
         4621    1
          Name: quantity, Length: 4622, dtype: int64
```

```
In [ ]:
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